

Final Dataset Validation Report

Customer Personality Segmentation | Module 5: Feature Engineering & Data Preprocessing (CPS-M05)

This is the last check before handing the dataset off to clustering. I verified the machine-learning-ready file (**customer_segmentation_ml_ready.csv**) against six criteria: missing values, duplicate records, data types, feature consistency, scaling completion, and encoding completion.

1. Dataset Overview

Metric	Value
Source dataset	marketing_campaign.csv (raw: 2,240 customers, 29 columns)
Rows after cleaning	2,237 (3 rows dropped — implausible birth years suggesting 100+ year old customers)
Final ML-ready columns	42 (1 ID column + 41 model-facing features)
Scaled numeric features	25 (RobustScaler)
One-hot encoded columns	16 (Marital_Status: 5, Preferred_Shopping_Channel: 5, Product_Preference: 6)

2. Validation Checklist

Check	Result	Status
Missing values	0 missing values across all 42 columns	PASS
Duplicate records (by ID)	0 exact duplicate rows	PASS
Data types	All 41 model-facing columns are numeric (int64/float64); ID retained as int64 for traceability	PASS
Feature consistency	Every categorical column was either label- or one-hot-encoded; no raw text columns remain in the dataset	PASS
Scaling completion	All 25 continuous/ordinal features scaled with RobustScaler; scaled means sit near 0 (avg. of col means is 0.02)	PASS
Encoding completion	16 one-hot columns present for Marital_Status, Preferred_Shopping_Channel, Product_Preference	PASS

3. A Note on Near-Duplicate Behavioral Profiles

While every row is a unique customer (unique ID, 0 exact duplicates), I did find that 184 customers share an identical *feature* profile with at least one other customer once IDs are excluded. This isn't a data quality issue — it happens naturally in a dataset this size when many customers share the same demographic bucket and simply spent \$0 in several product categories, producing identical scaled/encoded vectors. I left these rows in as-is, since they're legitimate distinct customers and removing them would bias the clustering toward pretending this group is smaller than it really is.

4. Scaling Sanity Check

A spot check across the 25 scaled columns confirms RobustScaler behaved as expected: column means sit close to 0 (the average of all scaled-column means is 0.02, well within tolerance) and no single column dominates the numeric range the way raw Income (tens of thousands) would have next to raw Total_Children (0-3). This confirms distance calculations in clustering won't be skewed by scale differences between features.

5. Conclusion

customer_segmentation_ml_ready.csv passes all six validation checks and is ready to be handed off to the clustering stage (K-Means, hierarchical clustering, DBSCAN, etc.). The companion file **customer_segmentation_full_engineered.csv** retains all intermediate columns (raw, engineered, encoded, transformed, and scaled) for anyone who wants to trace a feature back to its source or re-derive it differently later.